

Engineer Assistant — Post-Thesis Work Plan

Context

The Engineer Assistant thesis (76 pages, compiled) needs minor polishing before the **April 13** deadline. After that, the focus shifts to the May 8 presentation, potential IDETC paper revision (notification April 27), code cleanup, and system improvements — all the way through the IDETC conference at end of August.

Work schedule: - **Full-time (~40h/week)** until at least April 13, likely until May 8 - **Part-time (2 full days + 1 half day $\approx$ 20h/week)** from May 9 onwards

This gives roughly **560 usable hours** total. Since thesis polishing is minor, the full-time period before April 13 can also be used for code cleanup and early improvements. The plan below totals $\sim$ 484h.

Closed-loop manufacturing is out of scope. Everything else the user listed is included.

Fixed Milestones

Date	Event
Apr 3-6	Easter break (Good Friday – Easter Monday)
Apr 13	Thesis written report deadline
Apr 20-23	HPC AI conference (tentative, unavailable)
Apr 27	IDETC paper notification
May 1 (Fri)	Labour Day (Swiss holiday)
May 8	Final presentation (30 min)
May 14-15	Ascension Day + bridge day
May 19	IDETC final paper submission + author registration
May 25 (Mon)	Whit Monday (Swiss holiday)
Jun 29 – Jul 7	Conference + holidays (unavailable)
Aug 1 (Sat)	Swiss National Day (weekend, no impact)
Aug 23-26	IDETC conference

Phase 1: Thesis + Video + Early Cleanup (Mar 17 – Apr 13) ~120h (full-time)

Thesis content is largely final (mirrors the submitted IDETC paper). Polishing is light. Focus is: thesis, video, test fixes, and start deployment improvements.

Week	Task	Hours	Details
Mar 17-20 (Tue-Fri, 4d)	Thesis polish + YouTube video script	24h	Quick thesis read-through (content already solid from paper). Fix typos/formatting. Write video script and prepare demo scenarios.
Mar 23-27 (Mon-Fri, 5d)	Record + edit YouTube video	30h	Record demo: all agents, optimization, STL export, Prusa, RAG, Excalidraw. 5-8 min. Edit and upload. This is the main focus for the week.
Mar 30 – Apr 2 (Mon-Thu, 4d)	Fix tests + update deps	28h	Fix 2 pre-existing test failures (temperature assertion in <code>arxiv_agent</code> , <code>rag_agent</code>). Audit full test suite: verify correctness, remove obsolete tests, add missing coverage. Update dependencies one at a time with <code>pytest</code> after each. Add Python 3.12+ support (test compatibility, update Dockerfile, fix deprecations).
Apr 3-6	Easter break (Fri-Mon)	0h	No work.
Apr 7-10 (Tue-Fri, 4d)	Start deployment simplification	32h	Prerequisite: Ensure MMORE PR #240 is merged before MMORE integration changes. (1) MMORE as dependency: Replace external <code>~/Desktop/mmcore</code> folder with pip package or git submodule (depends on PR #240 merge). (2) Docker Compose profiles (<code>--profile full/dev/eval</code>). Replace <code>env-var</code> feature flags with profiles. (3) Docker secrets or auto-generated random credentials for DB (no plaintext).
Apr 13 (Mon, 1d)	Submit thesis	8h	Final compile, submit. Remove thesis and paper submodules from main repo (they're separate repos, no longer needed in the codebase). Prusa MCP submodule sync.

Deliverables: Thesis submitted. YouTube video published. Tests green. Dependencies updated. Deployment restructured (Docker profiles + MMORE).

Phase 2: Presentation + Standardization + arXiv (Apr 14 – May 7) ~120h (full-time)

Presentation is the main focus. Standardization and GUI launcher (deferred from Phase 1) fill the gaps.

Note: Apr 20-23 is blocked by the HPC AI conference (tentative). Presentation first draft must be done by Apr 17.

Week	Task	Hours	Details
Apr 14-17 (Tue-Fri, 4d)	Presentation first draft + rename to EngiAI	32h	~25-30 slides, 30 min talk. Structure: motivation → architecture → tool integration → benchmarks → demo video clip → conclusions. Reuse thesis figures. Must have a complete first draft by Apr 17. Rename repo from <code>engineer-assistant</code> to <code>EngiAI</code> (package name, imports, Docker images, README, docs, CI).
Apr 20-23	HPC AI conference (tentative)	0h	Unavailable.
Apr 24 (Fri, 1d)	Presentation refinement	8h	Polish slides post-conference. Fresh eyes review.
Apr 27-30 (Mon-Thu, 4d)	Standardization (EngiBench/EngiOpt style) + rehearsal	32h	Match EngiBench/EngiOpt structure exactly: conventional commits via <code>conventional-commit-lint</code> pre-commit hook, Ruff config (line-length 124, Google docstrings, <code>select = ["ALL"]</code>), Pyright, Codespell, dynamic version from <code>__init__.py</code> , <code>CITATION.cff</code> , <code>CONTRIBUTING.md</code> , organized pyproject.toml extras, GPL-3.0, same CI/CD pipeline (GitHub Actions: automated testing, linting, releases, changelog, Docker/PyPI publishing, GitHub Pages docs). Update Sphinx docs to match current features + new name. 1 rehearsal. Apr 27 = paper notification — if revision needed → see Phase 3a.
~~May 1 (Fri)~~	Labour Day (Swiss holiday)	0h	Off.
May 4-7 (Mon-Thu, 4d)	arXiv paper + rehearsals + start GUI launcher	32h	Convert ASME paper to arXiv format (strip class deps, self-contained figures, arXiv metadata). Upload preprint. 2 full timed rehearsals + Q&A prep. Backup plan: pre-recorded demo video. Start cross-platform GUI launcher (Python + customtkinter/PySide6, PyInstaller): start/stop Docker, show health, open browser. Admin panel in launcher (password-protected): API keys (encrypted, masked), Docker service config. Streamlit settings remain open : model selection, voice, theme — accessible to all users.
May 8 (Fri)	Presentation	8h	Present.

Deliverables: Presentation delivered. arXiv preprint submitted. Codebase standardized (EngiBench/EngiOpt style). GUI launcher + secure settings working.

Phase 3: System Improvements + Paper Revision (May 9 – Jun 27)

~110h (part-time, 20h/week)

Most code cleanup was done in Phases 1-2. This phase focuses on system improvements and paper revision.

3a. IDETC Paper Revision + Final Submission + GUI Launcher — May 9-19, ~20h (HARD DEADLINE)

Paper notification is Apr 27, final submission is **May 19**. This takes priority. GUI launcher finish runs in parallel.

Note: May 14 (Thu) = Ascension Day, May 15 (Fri) = bridge day (likely off).

Week	Task	Hours	Details
May 9-13 (Mon-Wed before Ascension, 3d)	Paper revision + finish GUI launcher	12h	Address reviewer feedback (if needed). Finish and test GUI launcher + admin panel from Phase 2.
~~May 14-15~~	Ascension + bridge day	0h	Off.
May 16-19 (Mon only before deadline)	Camera-ready + author registration	8h	Final formatting, submit by May 19 . Register as author.

If paper accepted without revision, these hours go to cleanup. Either way, the May 19 deadline must be met.

3b. Deep Code Cleanup — May 19 – Jun 6, ~35h

Week	Task	Hours	Details
May 19-23 (Mon-Fri)	Dead code removal + deduplication	10h	Audit codebase for unused imports, functions, dead code paths, and duplicate logic. Use ruff's <code>F401 / F841</code> . Check for duplicate patterns across agents. Remove stale files. Prune unused deps.
May 19-23	Prusa MCP cleanup + standardization	10h	Sync submodule (currently detached HEAD). Refactor and standardize to EngiBench/EngiOpt conventions: conventional commits, ruff/pyright config, CITATION.cff, CONTRIBUTING.md, pyproject.toml structure, GPL-3.0.
~~May 25~~	Whit Monday (Swiss holiday)	0h	Off.
May 26 – Jun 6 (Tue-)	(Extract Excalidraw component)	15h	Extract <code>src/ui/components/excalidraw/</code> to standalone <code>streamlit-excalidraw</code> repo with own CI.

Week	Task	Hours	Details
Fri + Mon-Fri, 9d)			Standardize to EngiBench/EngiOpt conventions. Do before MCP Excalidraw work (Phase 4).

3c. System Improvements — Jun 8 – Jun 26, ~55h

Week	Task	Hours	Details
Jun 8-12 (Mon-Fri)	Full Ollama support in chatbot	15h	Ollama model selection in Streamlit settings. Test with llama3, mistral. Handle tool-calling quirks. Add Ollama container to Docker Compose.
Jun 15-19 (Mon-Fri)	Better outputs for long-running tasks + Error recovery	20h	Streaming progress in Streamlit during optimization. Intermediate design visualization. HPC job polling with real-time status. Retry with exponential backoff for API calls. LangGraph error nodes. Graceful degradation (MMORE down → local, Prusa unreachable → save locally).
Jun 22-26 (Mon-Fri)	Experiment recording system + Google audio model support	20h	Set up recording infrastructure for human-AI experiments (Opencast/Tobira or OBS-based): screen capture + webcam, session storage/management. Gemini audio integration (1:1 with OpenAI voices, extend <code>voices.py</code>).

Jun 29 – Jul 7: Conference + Holidays (unavailable)

Phase 4: Feature Development (Jul 8 – Aug 15) ~120h (part-time, 20h/week)

4a. Core Features — Jul 8-24, ~55h

Week	Task	Hours	Details
Jul 8-10 + Jul 13-17 (8d)	Interactive MCP Excalidraw whiteboard	35h	Integrate the official excalidraw-mcp server with the existing Streamlit Excalidraw component (<code>src/ui/components/excalidraw/</code>). Add as a new MCP service (like Prusa MCP). Connect agent to whiteboard: agent can draw topology results, annotate designs; user sketches constraints that agent interprets. Bridge the MCP server ↔ Streamlit component communication.
Jul 20-24 (Mon-Fri)	More RAG file formats + Extend to other problems/ models	20h	Extend MMORE client: .pptx, .ipynb, video files, and other common formats. Update <code>rag_tools.py</code> . Containerized problem solvers (input/output JSON interface). Example container for new problem type. Update problem registry.

4b. New Agents + Repo Transfer — Jul 27 – Aug 14, ~65h

Week	Task	Hours	Details
Jul 27-31 + Aug 3 (Mon-Mon, 6d)	Self-writing training code agent	30h	New <code>TrainingAgent</code> (BaseAgent pattern). Tools: <code>generate_training_script</code> , <code>run_training</code> , <code>evaluate_model</code> . Automatic validation: generated code is compiled/syntax-checked before submission, no manual review step. HPC integration for Euler. Enable extended HPC benchmarks (longer training runs, multi-model evaluation). Full testing: unit tests for agent/tools, integration test with mock HPC, end-to-end test on Euler.
Aug 4-5 (Tue-Wed)	Repo transfer to lab GitHub org	8h	Repo already renamed to EngiAI (Phase 2). Clean any remaining hardcoded paths/credentials. Final verification of EngiBench/EngiOpt-style structure. Transfer/fork under lab org. Handoff document. Tag v2.0.0.
Aug 6-7 (Thu-Fri)	New computer setup	10h	Set up dev environment on new machine. Verify Docker deployment from clean clone (serves as deployment validation). Install deps, test all services via GUI launcher.
Aug 10-14 (Mon-Fri)	Buffer / overflow	17h	Catch up on any tasks that slipped. Polish features. Final testing.

Phase 5: IDETC + Wrap-Up (Aug 17 – Aug 28) ~20h

Date	Task	Hours
Aug 17-21 (Mon-Fri)	IDETC prep + travel	10h
Aug 23-26 (Sun-Wed)	IDETC conference	—
Aug 27-28 (Thu-Fri)	Final wrap-up	10h

Wrap-up: update README/docs with all new features, final release tag, archive, status document for the lab.

Hours Summary

Phase	Period	Pace	Hours	Focus
1. Thesis + Video + Tests	Mar 17 – Apr 13	Full-time	122h	Thesis, video, test fixes, deps, deployment restructure
2. Presentation + Standardization + arXiv	Apr 14 – May 8	Full-time	112h	Slides, rename, standardization, CI/CD, arXiv, GUI launcher (minus HPC AI conf + May 1)

Phase	Period	Pace	Hours	Focus
3. Cleanup + Improvements	May 9 – Jun 27	Part-time	110h	Paper revision, dead code, Excalidraw extraction, Prusa MCP, Ollama, outputs, errors, recording, audio
4. Features	Jul 8 – Aug 15	Part-time	120h	MCP whiteboard, training agent, RAG formats, containers, repo transfer, new computer
5. IDETC	Aug 17 – Aug 28	—	20h	Conference + wrap-up
Total			484h	

Available hours: - Full-time (Mar 17 – May 8, ~7 weeks minus Easter/HPC AI conf): ~230h - Part-time (May 9 – Aug 31, ~16 usable weeks × 20h): ~320h - **Total available: ~550h → buffer of ~66h**

Is This Feasible?

Yes, comfortably. The full-time period (Phases 1-2) handles all the deadline-critical work plus most code cleanup. By May 8, the thesis is submitted, presentation done, arXiv uploaded, docs updated, and the codebase is clean. Phases 3-4 at part-time are modular — individual tasks can be paused, reordered, or dropped if paper revision demands more time.

If IDETC paper is accepted without revision, you gain ~30h extra buffer.

Risk Factors

Risk	Impact	Mitigation
Paper rejected (Apr 27)	Frees ~30h; need to decide on resubmission venue	arXiv preprint ensures visibility. Redirect hours to features.
Heavy paper revision	Eats into Phase 3b cleanup time	Defer some cleanup tasks to Phase 4.
MCP Excalidraw harder than expected	30h may not be enough	Ship a simpler version (display-only, no bidirectional drawing) and document full vision.
Ollama models can't do tool calling	Feature limited in scope	Document which models are compatible. Fall back to "Ollama for Q&A, API for tools."
Dependency updates break things	Could cascade	Update one at a time, test after each. Keep pre-update tag.

Optional / Deferred Tasks

These can be scheduled if buffer hours open up (e.g., paper accepted without revision):

- **Ablation studies** (prompt/temperature/model tests) — ~15-20h. May be useful for arXiv paper or future publication. Run if paper reviewers request it or if time permits.

Tasks NOT Included

- **Closed-loop manufacturing feedback** — out of scope per user decision.
- **User studies** — mentioned in thesis as future work, not planned here.